

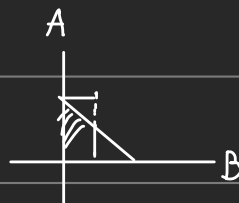
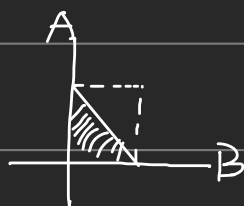
一、

$$1. X(1) = A+B \leq 1$$

$$P\{X(1) \leq 1\} = \frac{1}{2}$$

$$P\{X(1) \leq 1 | X(0) \leq 0.5\} = \frac{P(A+B \leq 1, B \leq 0.5)}{P(B \leq 0.5)}$$

$$= \frac{\frac{3}{8}}{\frac{1}{2}} = \frac{3}{4}$$



$$R_X(t,s) = E[(A_t+B)(A_s+B)]$$

$$= E[A^2ts + ABs + ABt + B^2]$$

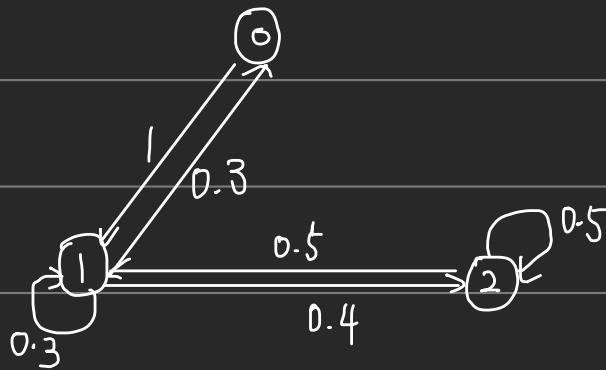
$$= (\int_0^1 x^2 dx) ts + \frac{1}{4}(s+t) + \int_0^1 x^2 dx$$

$$= \frac{1}{3}ts + \frac{1}{4}(s+t) + \frac{1}{3}$$

$$2. R_X(\tau) = \frac{1}{4}e^{-|\tau|} + \frac{1}{12}e^{-3|\tau|}$$

$\therefore S_X(\omega)$ 在 $\omega=0$ 处没有冲激项, 故 $\mu_X=0$

3.



$$f_{22}^{(3)} = 0.5 \times 0.3 \times 0.4 = 0.06$$

$$\begin{array}{r} 0.06 \\ 8.125 \\ 0.2 \end{array}$$

$$p_{22}^{(3)} = 0.5 \times 0.3 \times 0.4 + (0.5)^3 + 0.5 \times 0.5 \times 0.4$$

$$= 0.385$$

$$P\{X_2=2\} = \frac{1}{2} \times p_{02}^{(2)} + \frac{1}{2} \times p_{12}^{(2)}$$

$$P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0.3 & 0.3 & 0.4 \\ 0 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0.3 & 0.3 & 0.4 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

$$p_{02} = 0.4 \quad p_{12} = 0.32 \quad P\{X_2=2\} = 0.36$$

$$P\{X_0=1, X_2=2, X_3=2\} = \frac{1}{2} \times 0.32 \times 0.5 = 0.08$$

4. $P\{N(2)=2 | N(1)=1\} = P(N(2)-N(1)=1)$

$$N(2)-N(1) \sim P\left(\int_1^2 2dt\right) = P(2)$$

$$\therefore P(N(2)-N(1)=1) = \frac{e^{-2} 2^1}{1!} = 2e^{-2}$$

$$P\{N(3)=3 | N(1)=1\} = P(N(3)-N(1)=2)$$

$$N(3) - N(1) \sim P\left(\int_1^2 2dt + \int_2^3 1dt\right) = P(3)$$

$$\therefore P(N(3) - N(1) = 2) = \frac{e^{-3} \times 3^2}{2!} = 4.5e^{-3}$$

$$\text{二、(1) } \text{Cov}(B(1), B(2)) = \min\{1, 2\} = 1$$

$$(2) B(2) \sim N(0, 2)$$

$$f(x) = \frac{1}{\sqrt{2\pi} \sqrt{2}} e^{-\frac{x^2}{2 \times 2}} = \frac{1}{2\sqrt{\pi}} e^{-\frac{x^2}{4}}$$

(3) 是题干有误

$$(4) P(B(1) \geq 1 | B(2) = 2)$$

$$\tilde{B}(t) = \begin{cases} tB(1/t), & t > 0 \\ 0, & t = 0 \end{cases}$$

$$\therefore P(B(1) \geq 1 | B(2) = 2) = P(\tilde{B}(1) \geq 1 | \tilde{B}(\frac{1}{2}) = 1) = \frac{1}{2}$$

$$\begin{aligned} (5) E(B(2)B(4)B(6)) &= E[B(2)[B(4) - B(2) + B(2)][(B(6) - B(4) + B(4) + B(2))]] \\ &= E[B(2)^2((B(6) - B(2)) + B(2))] = E[B(2)^3] \end{aligned}$$

$\therefore B(2) \sim N(0, 2)$, 其奇数阶中心矩为0

$$\text{三、(1) } N(t) = N_1(t) + N_2(t) \quad E[N(t)] = 5$$

$$(2) P(N(1) = 0, N(2) - N(1) \geq 1)$$

$$= e^{-5} \times [1 - e^{-5}] = e^{-5} - e^{-10}$$

$$(4) \frac{P(N(1)=1, N(2) \leq 2)}{P(N(2) \leq 2)}$$

$$= \frac{e^{-2} \times 1 \times e^{-3} \times 3 \times [e^{-5} + e^{-5} \frac{5}{1}]}{e^{-5 \times 2} + e^{-5 \times 2} \frac{(5 \times 2)^1}{1!} + e^{-5 \times 2} \frac{(5 \times 2)^2}{2!}}$$

$$= \frac{18}{61} \quad (\text{第一小时恰好钓到一条鲤鱼, 那})$$

$$\text{鳊鱼会不会有, 有则是 } \frac{24}{61})$$

四、(1) $E[X(t)] = 0.5 E[\sin(t-A)]$

$$= 0.5 E[\sin(t) \cos A - \cos t \sin A] = 0$$

$$R_X(t, t+\tau) = E[X(t) X(t+\tau)]$$

$$= E[B^2 \sin(t-A) \sin(t+\tau-A)]$$

$$= 0.5 \times E[\frac{1}{2} \cos(\tau) - \frac{1}{2} \cos(2t+\tau-2A)]$$

$$= \frac{1}{4} \cos(\tau)$$

$\therefore$ 均值函数为常数, $R_X(t, t+\tau)$ 为 τ 的函数, 为宽平稳过程

$$(2) S_X(\omega) = \frac{1}{4} F[\cos(t)] = \frac{\pi}{4} [\delta(\omega-1) + \delta(\omega+1)]$$

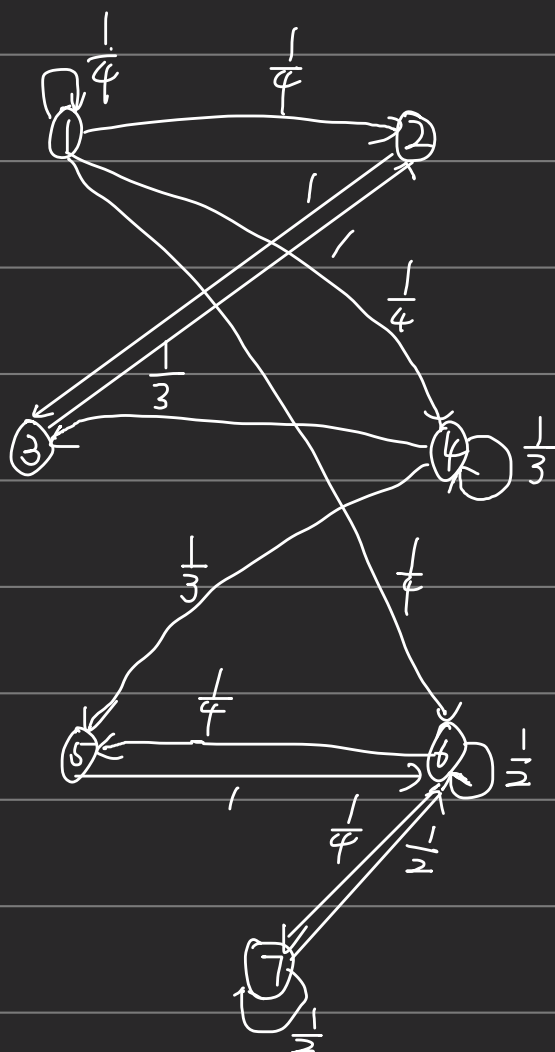
$$(3) \langle X(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T X(t) dt = 0$$

$$\langle X(t) X(t+\tau) \rangle = \lim_{T \rightarrow \infty} \int_0^T X(t) X(t+\tau) = \frac{B^2}{2} \cos \tau$$

(4) 并不是各态历经过程, 因为 $\frac{B^2}{2}$ 几乎处处不等于 $\frac{1}{4}$

(5) ?

五、(1)



$\{1, 4\}, \{2, 3\}, \{5, 6, 7\}$

其中 $\{2, 3\}, \{5, 6, 7\}$ 是闭的。

(2) $d_1=1$ $d_2=d_3=2$ $d_4=1$ $d_5=d_6=d_7=1$

1, 4 暂留, 2, 3, 5, 6, 7 正常返

(3) 对于 $\{2, 3\}$ 这个马尔可夫链

$$(\pi_2 \ \pi_3) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = (\pi_2 \ \pi_3) \Rightarrow \begin{cases} \pi_2 = \frac{1}{2} \\ \pi_3 = \frac{1}{2} \end{cases}$$

$$\therefore \mu_2 = \mu_3 = 2$$

$$\text{又对 } \{5, 6, 7\}$$

$$(\pi_5, \pi_6, \pi_7) \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = (\pi_5, \pi_6, \pi_7)$$

$$\Rightarrow \begin{cases} \pi_5 = \frac{1}{7} \\ \pi_6 = \frac{4}{7} \\ \pi_7 = \frac{2}{7} \end{cases}$$

$$\therefore \mu_5 = 7, \quad \mu_6 = \frac{7}{4}, \quad \mu_7 = \frac{7}{2}$$

$$\begin{aligned} (4) \quad P\{T_{\{5,6,7\}} < T_{\{2,3\}}\} &= \frac{1}{2} + P(X_0 = 1)P\{T_{\{5,6,7\}} < T_{\{2,3\}}\} \\ &= \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} \end{aligned}$$

$$(5) \quad \lim_{n \rightarrow +\infty} P(X_n = 1) = 0$$

$$\lim_{n \rightarrow +\infty} P(X_n = 6) = \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{1}{2} \times \frac{4}{7} = \frac{3}{7}$$

